# Jason Li

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## EDUCATION

**University of Utah** Salt Lake City, UT
*Ph.D. in Computing, Data Management and Analysis Track* *Aug. 2025 – Present*
- GPA: 4.0/4.0

**University of Florida** Gainesville, FL
*B.S. in Computer Science and Statistics* *Aug. 2021 – May 2025*
- GPA: 3.94/4.0; GRE: 337 (Q: 170, V: 167)
- Honors: UF Honors Program, Undergraduate Research Scholars Program, Benacquisto Scholar (Full-Ride Scholarship), Dean's List, National Merit Finalist

## PUBLICATIONS

1. **Jason Li**, Rodrigo de la Nuez Moraleda, Dhruv Meduri, Kelin Zia, and Bei Wang, "Explainable Reaction Atlas," *IEEE Workshop on Topology meets Artificial Intelligence (TopoInVis Connect), IEEE VIS*, 2026. **Accepted.**

2. Andres Espinosa, Cody Flynn, **Jason Li**, Han Mach, Maximillian Banach, Cathy Quan, Jorg Peters, and Brian Stensrud, "Multi-Agent Board Game Strategy Through Simulation," *Interservice/Industry Training, Simulation and Education Conference (I/ITSEC)*, 2025. **Best Paper Award.**

## AWARDS

- **I/ITSEC 2025 Best Paper Award** — *Multi-Agent Board Game Strategy Through Simulation*

## RESEARCH EXPERIENCE

**Research Assistant** Dec. 2025 – Present
*Topological Intelligence Lab (TILab), University of Utah* *Salt Lake City, UT*
- Conduct research under Ph.D. advisor Professor Bei Wang Phillips at the intersection of explainable AI, deep learning, and data visualization.
- Develop methods for visualizing and interpreting latent representations learned by deep learning models to improve model transparency and understanding.
- Investigate explainable AI techniques and human-AI interaction methods for enabling users to explore, interpret, and reason about complex model behavior.

**Research Assistant** Feb. 2024 – May 2024
*Professor Leo Duan's Research Group, University of Florida* *Gainesville, FL*
- Statistically analyzed and visualized travel movement trends caused by Hurricane Ian using R.
- Identified traffic-volume patterns before, during, and after Hurricane Ian.
- Parsed and cleaned census-tract-level geospatial data using Python.

**Research Assistant** Aug. 2023 – Jan. 2024
*Embodied Learning & Experience Lab, University of Florida* *Gainesville, FL*
- Developed a smartwatch application to facilitate learning in everyday life for children ages 9–12.
- Reviewed and synthesized peer-reviewed literature to inform research initiatives.

## PROFESSIONAL & TEACHING EXPERIENCE

**Teaching Assistant** Aug. 2024 – May 2025
*University of Florida* *Gainesville, FL*
- Tutored students in Information and Database Systems.
- Graded student submissions, administered examinations, and presented lecture material during discussion sections.

**Software Engineer Intern** Jun. 2024 – Aug. 2024
*Fidelity Investments* *Salt Lake City, Utah*
- Developed a secure web-based funds-transfer platform for digital banking using Angular and GraphQL.
- Implemented backend API services to query and format data and support front-end navigation.

**Peer Tutor** Oct. 2023 – May 2024
*Academic Tutoring Center, University of Florida* *Gainesville, FL*

- Provided one-on-one and small-group tutoring in the core programming curriculum for Computer Science and Computer Engineering students.

**IT Department Business Analyst Intern** May 2022 – Aug. 2022
*Supplemental Health Care* *Sandy, Utah*

- Automated user-permission assignment, reducing user creation time by 50%.
- Implemented a notification system supporting upskilling for more than 10,000 affiliated nurses.
- Collaborated with developers, stakeholders, project managers, and QA auditors throughout SDLC processes.

## PROJECTS

**Explainable Reaction Atlas** 2026

- Developed an LLM-based explanation framework for interpreting chemical reaction latent spaces represented as interactive tree-structured atlases.
- Implemented Neighborhood and Path Explainers to summarize local reaction regions, compare branches, and analyze transitions between user-selected reactions.
- Evaluated the framework on the Schneider 50K dataset of approximately 50,000 chemical reactions, enabling explanation-guided exploration of reaction patterns and locally inconsistent regions.

**Multi-Agent Board Game Strategy Through Simulation** 2025

- Developed a custom simulation environment for the turn-based strategy game *Settlers of Catan* to support reinforcement-learning experiments.
- Implemented a Deep Q-Learning agent in PyTorch using an epsilon-greedy exploration strategy.
- Trained and evaluated the agent against a state-of-the-art benchmark, achieving a 64% win rate in one-vs-one simulations.

**Household Location Choice Model**

- Modeled housing choice behavior in Salt Lake County, Utah, reducing the feature set from 150 variables to 12.
- Applied stepwise regression, LASSO, principal component analysis, random forests, variable transformations, interactions, and genetic-algorithm optimization.

**Online Shopping Application**

- Built an interactive full-stack shopping application using React, Ant Design, Spring MVC, Hibernate, MySQL, and Node.js.
- Developed REST APIs for registration, search, and checkout and deployed the application using Amazon EC2 and RDS.

**Online Room Booking Site**

- Developed a React and Spring Boot room-booking application with reservation, search, upload, and deletion services.
- Implemented MySQL persistence, Google Cloud Storage, Elasticsearch geospatial search, and JWT authentication.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, Java, C++, C#, SQL, R, MATLAB, Kotlin

**AI/ML & Scientific Computing:** PyTorch, TensorFlow, scikit-learn, Hugging Face, NumPy, pandas, Jupyter, RAG, OpenAI API, Model Context Protocol (MCP), Pydantic AI, vLLM

**Backend & Data Systems:** FastAPI, Node.js, Spring Boot, Spring MVC, .NET Framework, Hibernate, GraphQL, REST APIs, MySQL, Elasticsearch, Apache Tomcat

**Frontend, Mobile & Visualization:** React, Angular, Svelte, HTML/CSS, D3.js, Android

**Cloud & Infrastructure:** Docker, Docker Compose, Caddy, AWS EC2, Amazon RDS, Oracle Cloud Infrastructure, Google App Engine, Google Cloud Storage, Ubuntu/Linux, DNS, HTTPS/TLS

**Developer Tools & Practices:** Git/GitHub, Pytest, Distributed Computing, Multithreading, Agile (Scrum/Kanban), Jira